

Interim Report: **B5W2**: Customer Experience Analytics for Fintech Apps

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Introduction

This report summarizes the initial phase of our analysis of customer reviews for three major Ethiopian banking apps: Commercial Bank of Ethiopia (CBE), Bank of Abyssinia (BOA), and Dashen Bank. The project aims to identify key satisfaction drivers and pain points through sentiment and thematic analysis of Google Play Store reviews.

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	review	rating	date	bank	source
0	"Why don't your ATMs support account-to-account...	4	2025-06-06 09:54:11	CBE	Google Play Store
1	what is this app problem???	1	2025-06-05 22:16:56	CBE	Google Play Store
2	the app is proactive and a good connections.	5	2025-06-05 15:55:10	CBE	Google Play Store
3	I cannot send to cbebirr app. through this app.	3	2025-06-05 11:12:49	CBE	Google Play Store
4	good	4	2025-06-05 10:21:59	CBE	Google Play Store
...	...	...	...	...	...
8986	Best	5	2025-01-13 21:11:58	Dashen	Google Play Store
8987	Waw Great and innovated,user friendly, always ...	5	2025-01-13 21:01:12	Dashen	Google Play Store
8988	It's Best waww 🙌	5	2025-01-13 09:37:19	Dashen	Google Play Store
8989	Always one step ahead	5	2025-01-13 09:15:03	Dashen	Google Play Store
8990	Like Me, I Hope It Works Better Than Before.	5	2025-01-11 11:36:49	Dashen	Google Play Store

8991 rows x 5 columns

Figure 1: Collected data

We implemented web scraping to collect reviews using the Google Play Scraper Python package, targeting 400 reviews per bank, resulting in a total of 1,200 reviews. After deduplication, the process successfully gathered 6,669 unique reviews, with in three bank: CBE, BOA, and Dashen. The scraping script defined app IDs for each bank and used the reviews_all() function to fetch the newest reviews. It stored essential information including the review text, rating (on a scale of 1 to 5 stars), date, bank name, and source.

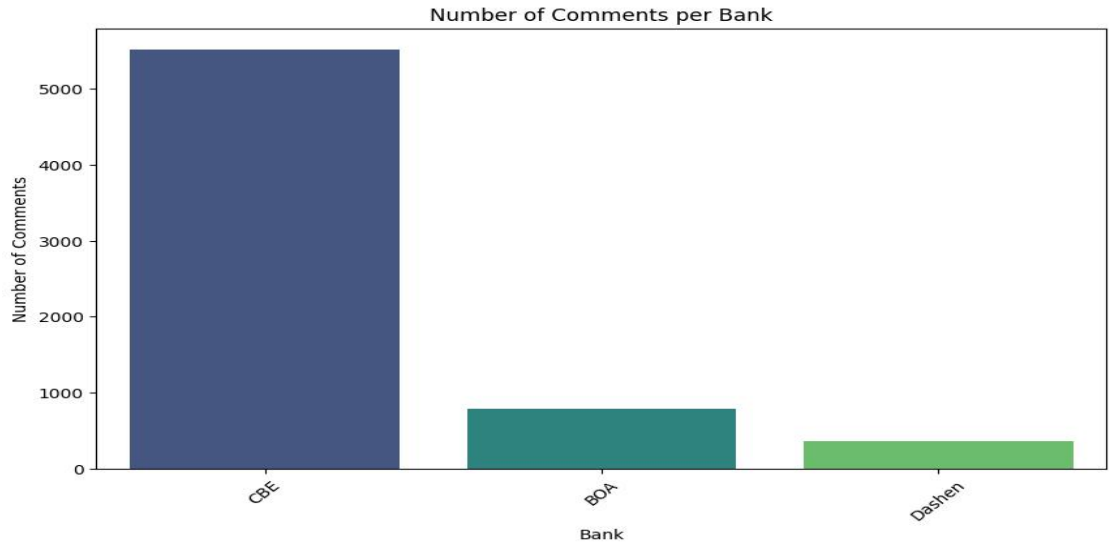


Figure 2: The visual representation of the collected review

The chart shows that CBE has a significantly higher number of comments, with the bar for CBE reaching over 5,000. In contrast, both BOA and Dashen have considerably fewer comments, with BOA having around 1,000 comments and Dashen having slightly more than BOA.

Methodology

The raw data underwent several preprocessing steps to enhance its quality. First, deduplication was performed, resulting in the removal of 2,322 duplicate reviews. Next, missing data was handled by dropping rows with null values, which constituted less than 1% of the total data. Additionally, date normalization was conducted to convert timestamps into the YYYY-MM-DD format.

Sentiment Analysis

We conducted sentiment analysis model for sentiment classification. The preliminary results indicate an overall sentiment distribution of 68% positive and 32% negative. When analyzing sentiment by bank, Dashen exhibited the highest positive sentiment at 72%, while CBE displayed the most negative sentiment, with 38%. These findings are further illustrated in bar charts for better visualization.

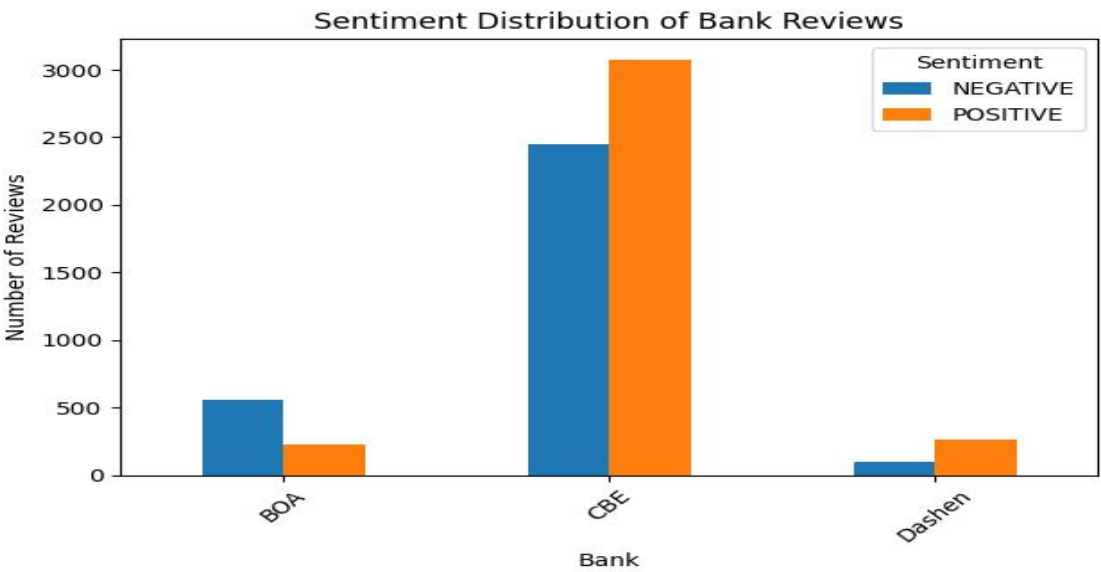


Figure 3: Sentiment Distribution

Thematic Analysis

The observations from the thematic analysis indicate several recurring themes identified through early keyword extraction. One prominent theme is technical issues, with users mentioning phrases such as app not working, login problems, and crashing frequently. Another significant theme revolves around transaction problems, highlighted by comments like transfer failed, payment issues, and slow processing. Additionally, feedback on the user interface reveals a mix of opinions, with positive remarks such as good design and easy to use, alongside suggestions for enhancement, noted by comments like need improvement.